

SALEHA MUZAMMIL

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EDUCATION

University of Virginia

PhD in Computer Science

GPA: 4.0/4.0

Aug. 2025 – May 2030 (Expected)

Virginia, USA

Coursework till date: Serverless AI, Data Privacy, Security for AI Systems, Constrained Aware Generative-AI, Probabilistic Machine Learning

National University of Computer and Emerging Sciences

Bachelor of Science in Computer Science

Cum Laude Distinction 3.67/4.0, ranked 7/231 students

Sep. 2020 – June 2024

Lahore, Pakistan

PUBLICATIONS

Private Yet Accurate: A Decentralized Approach to System Intrusion Detection

Feb. 2026

Jinghan Zhang, Sharon Bijju, *Saleha Muzammil*, Wajih Ul Hassan

- Workshop paper in PRISM, NDSS 2026. Designed a federated learning system for intrusion detection that preserves privacy while matching centralized model accuracy, reducing network costs by 170× compared to existing approaches.

Source Code Hotspots: A Diagnostic Method for Quality Issues

April 2026

*Saleha Muzammil**, Mughees Ur Rehman, Zoe Kotti, Diomidis Spinellis

- Accepted at MSR (Mining Software Repositories) 2026. Performed a thorough empirical analysis of source code hotspots and analyzed the impact of automation bots on code churn.

Towards Small Language Models for Security Query Generation in SOC Workflows

Dec. 2025

*Saleha Muzammil**, Rahul Reddy*, Vishal Kamalakrishnan, Hadi Ahmadi, Wajih Ul Hassan

- Accepted at RAID 2026. Designed and evaluated a cost-effective small language model architecture combining LoRA fine-tuning, error-aware prompting, and LLM-based refinement for enterprise-scale cybersecurity query translation across Microsoft Defender and Sentinel platforms.

Towards Fairer AI: Multi-Agent Debiasing of LLMs With Online Evidence Retrieval

Nov. 2025

Mughees Ur Rehman*, *Saleha Muzammil**

- Accepted at AAAI Symposium 2025 in the AI for Social Good track. Designed a multi-agent system to debias LLM reasoning via online evidence retrieval, involving large-scale dataset curation, statistical evaluation, and error analysis.

EXPERIENCE

CISPA Helmholtz Center for Information Security

June 2026 – August 2026

Research Intern

- Worked with Dr. Michael Pradel on creating an end-to-end evasive pipeline to make AI coding agents trip, paper to be submitted to IEEE S&P 2027 in November.

University of Virginia

Sep. 2025 – Present

Graduate Research Assistant

- Working under the supervision of Wajih Ul Hassan on system security for AI agents. Worked on a project focused on forensic root-cause attribution of AI agents using their execution traces. Paper has submitted to USENIX Security 2027.

Kaleidoscope

Dec. 2024 – Aug. 2025

Software Engineer

- Developed data pipelines and APIs enabling real-time analytics dashboards. Delivered 12 features that drove a \$1.2M/year revenue deal. Created serverless GraphQL APIs using AWS Appsync and Lambda with minimal response time. Developed and configured blueprints, analyzers, and widgets for data collection and visualization. Took part in SRE rotations to resolve production issues and eliminated high and critical software BOM vulnerabilities.

UIUC+ Summer Research Program

April 2024 – Dec. 2024

Research Intern

- Worked on provenance for software reproducibility under the supervision of Darko Marinov. My work included porting the system to macOS through interposition, TCP/HTTPS interception, and performance benchmarking. Conducted micro- and macro-benchmarking for performance analysis and implemented a CI workflow to uphold codebase quality. The work was submitted to ICSE'26.

DGlobal

June 2024 – Nov. 2024

Software Engineer

- Worked as a Full Stack Engineer at DGlobal, developing core systems for its digital banking platform to streamline customer modifications and product management using TypeScript, GraphQL, ReactJS, and NestJS.

Center of Speech and Language Technologies

June 2023 – March 2024

External Research Assistant

- Worked with Dr. Agha Ali Raza and Dr. Awais Athar, enhancing the Kahaani project, a speech-driven Urdu language platform.
- Developed BaatBot, a Urdu voice chatbot using ASR, TTS, and LLM systems, gaining expertise in Generative AI. Identified and pursued a research gap to improve child ASRs for the Urdu language.

HONORS & AWARDS

- Outstanding Graduate Service Award 2026 by Department of Computer Science, University of Virginia.
- CRA-WP Grad Cohort Workshop 2026 attendee.
- NSF-AAAI travel grant awardee to attend the AAI Symposium 2025.
- Finalist AI Challenge Pakistan 2024 (Top 3 among 100+ projects).
- Placed on the university's Dean List for 2020-2024. Bronze Medalist NUCES Spring 2024. (Undergraduate)
- Runner-Up in SOFTEC 2024 Software Project Competition.

TECHNICAL SKILLS

Programming: Python (Pandas, Numpy, Seaborn, PyTorch, Matplotlib, Scrapy, PySpark), C, C++, JavaScript, Typescript, GoLang, R

Databases: GraphQL, MySQL, MongoDB, Firebase, Hadoop (HDFS), Spark

Frameworks: ReactJS, NestJS, NodeJS, ExpressJS, Flask, Django

Developer Tools: GitHub, Git, AWS, Docker, Kubernetes

SERVICES

University of Virginia- CS Department

Jan. 2026 – Present

Graduate Experience Chair

- Organizing social events, research symposiums, seminars, and sessions for MS and PhD students to make their graduate life more productive and fun.

University of Virginia's Institute for Practical Ethics

Aug. 2025 – Present

Student Leader for Responsible Conduct of Research

- Led two discussion-based workshops on the “responsible conduct of research” for the Computer Science department over the course of the 2025-2026 academic year. The goal was to explore, in a discipline-specific way, what graduate students experience as ethically challenging in their professional culture.

TEDxFASTLahore

Aug. 2023 – Aug.2024

Licensee and Curator

- Got the license for FAST Lahore to organize TEDx events in the university. Organized and curated the first TEDx event for the university on the theme “Power of Manifestation” with the TED team.

Career Counseling Society, NUCES

Aug. 2021 – Aug.2023

General Secretary

- Organized two annual Job Fairs for the university, more than 50 recruitment events, speaker events and sessions.